



## Neet Test Hub Premium Test 09 Answer Key

### Physics

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 1 - C  | 2 - A  | 3 - A  | 4 - A  | 5 - C  | 6 - D  | 7 - D  | 8 - B  | 9 - A  | 10 - B |
| 11 - A | 12 - C | 13 - A | 14 - B | 15 - A | 16 - C | 17 - A | 18 - A | 19 - A | 20 - D |
| 21 - B | 22 - C | 23 - C | 24 - C | 25 - A | 26 - D | 27 - D | 28 - C | 29 - D | 30 - A |
| 31 - C | 32 - D | 33 - C | 34 - B | 35 - B | 36 - D | 37 - A | 38 - D | 39 - D | 40 - D |
| 41 - C | 42 - C | 43 - A | 44 - D | 45 - C |        |        |        |        |        |

### Chemistry

|        |        |        |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 46 - B | 47 - B | 48 - B | 49 - B | 50 - D | 51 - B | 52 - D | 53 - B | 54 - D | 55 - B |
| 56 - C | 57 - C | 58 - C | 59 - A | 60 - D | 61 - A | 62 - A | 63 - D | 64 - A | 65 - A |
| 66 - A | 67 - A | 68 - A | 69 - D | 70 - B | 71 - D | 72 - A | 73 - A | 74 - A | 75 - A |
| 76 - A | 77 - C | 78 - C | 79 - A | 80 - C | 81 - B | 82 - D | 83 - B | 84 - D | 85 - D |
| 86 - C | 87 - A | 88 - B | 89 - A | 90 - A |        |        |        |        |        |

### Biology - ( Botany )

|         |         |         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 91 - C  | 92 - C  | 93 - C  | 94 - D  | 95 - B  | 96 - B  | 97 - C  | 98 - D  | 99 - A  | 100 - A |
| 101 - C | 102 - A | 103 - A | 104 - C | 105 - C | 106 - C | 107 - D | 108 - B | 109 - B | 110 - C |
| 111 - C | 112 - A | 113 - D | 114 - D | 115 - B | 116 - D | 117 - B | 118 - B | 119 - D | 120 - C |
| 121 - B | 122 - B | 123 - C | 124 - C | 125 - A | 126 - D | 127 - C | 128 - B | 129 - B | 130 - D |
| 131 - D | 132 - A | 133 - A | 134 - D | 135 - D |         |         |         |         |         |

### Biology - ( Zoology )

|         |         |         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 136 - D | 137 - C | 138 - A | 139 - C | 140 - C | 141 - A | 142 - C | 143 - B | 144 - C | 145 - A |
| 146 - B | 147 - A | 148 - D | 149 - D | 150 - A | 151 - D | 152 - D | 153 - B | 154 - D | 155 - B |
| 156 - D | 157 - B | 158 - D | 159 - C | 160 - A | 161 - A | 162 - C | 163 - C | 164 - A | 165 - C |
| 166 - D | 167 - C | 168 - B | 169 - A | 170 - B | 171 - A | 172 - A | 173 - A | 174 - C | 175 - D |
| 176 - C | 177 - D | 178 - C | 179 - C | 180 - A |         |         |         |         |         |

## Physics

1. If masses of two point objects are tripled and distance between them is doubled, then gravitational force of attraction between them will ..... %

- A) Increase by 225%  
 B) Decrease by 56%  
 C) Increase by 125%  
 D) Decrease by 144%

**Solution : (Correct Answer: C)**

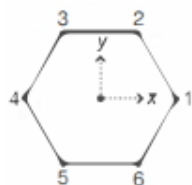
(c)

$$F = \frac{GMm}{R^2}$$

If masses are tripled and distance is doubled,

$$\begin{aligned} F' &= \frac{G \times 3M \times 3m}{4R^2} \\ &= \frac{9}{4} \frac{GMm}{R^2} \\ &= \frac{9}{4} F \end{aligned}$$

2. Six objects are placed at the vertices of a regular hexagon. The geometric centre of the hexagon is at the origin with objects 1 and 4 on the  $X$ -axis (see figure). The mass of the  $k$  th object is  $m_k = k^i M |\cos \theta_k|$ , where  $i$  is an integer,  $M$  is a constant with dimension of mass and  $\theta_k$  is the angular position of the  $k$  th vertex measured from the positive  $X$ -axis in the counter-clockwise sense. If the net gravitational force on a body at the centroid vanishes, the value of  $i$  is



- A) 0      B) 1      C) 2      D) 3

**Solution : (Correct Answer: A)**

(a)

For a mass  $m$  at centroid of hexagon (at origin), net force is zero when  $\Sigma F_x = 0$  and  $\Sigma F_y = 0$

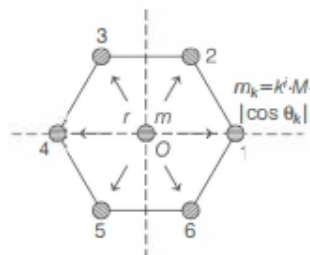
Now,  $\Sigma F_x =$  sum of all  $x$ -components of forces on  $m$  due to masses at vertices of hexagon.

$$\begin{aligned} &= \frac{Gm}{r^2} (\Sigma (k^i M |\cos \theta_k| \cdot \cos \theta_k)) \\ &= \frac{GMm}{r^2} (1^i \cdot |\cos 0^\circ| \cdot \cos 0^\circ + 2^i |\cos 60^\circ| \cdot \cos 60^\circ \\ &\quad + 3^i |\cos 120^\circ| \cdot \cos 120^\circ + 4^i |\cos 180^\circ| \cdot \cos 180^\circ \\ &\quad + 5^i |\cos 240^\circ| \cdot \cos 240^\circ + 6^i |\cos 300^\circ| \cdot \cos 300^\circ) \\ &= \frac{GMm}{r^2} \cdot \left(1^i + \frac{2^i}{4} - \frac{3^i}{4} - 4^i - \frac{5^i}{4} + \frac{6^i}{4}\right) \end{aligned}$$

As  $\Sigma F_x = 0$ , for net force on  $m$  to be zero, we have

$$1^i + \frac{2^i}{4} - \frac{3^i}{4} - 4^i - \frac{5^i}{4} + \frac{6^i}{4} = 0$$

Above equation is satisfied with  $i = 0$ .



3. If the mass of the planet is 10% less than that of the earth and radius is 20% greater than that of earth, the acceleration due to gravity on the planet will be

- A)  $\frac{5}{8}$  times that on the surface of earth  
 B)  $\frac{3}{4}$  times that on the surface of earth  
 C)  $\frac{1}{2}$  times that on the surface of earth  
 D)  $\frac{9}{10}$  times that on the surface of earth

**Solution : (Correct Answer: A)**

$$\begin{aligned} g' &= \frac{G(\frac{9}{10}M)}{(\frac{12}{10}R)^2} \\ &= \frac{5}{8} \frac{GM}{R^2} \\ &= \frac{5}{8} g \end{aligned}$$

4. Let  $\omega$  be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles. An object weighed at the equator gives the same reading as a reading taken at a depth  $d$  below earth's surface at a pole ( $d \ll R$ ). The value of  $d$  is

- A)  $\frac{\omega^2 R^2}{g}$       B)  $\frac{\omega^2 R^2}{2g}$   
 C)  $\frac{2\omega^2 R^2}{g}$       D)  $\frac{\sqrt{Rg}}{g}$

**Solution : (Correct Answer: A)**

Lets  $\omega =$  the angular velocity of the earth's rotation about its axis.

$g =$  acceleration due to gravity.

$R =$  radius of earth

$d =$  depth a below earth surface at which the acceleration due to gravity is same at the equator and the poles.

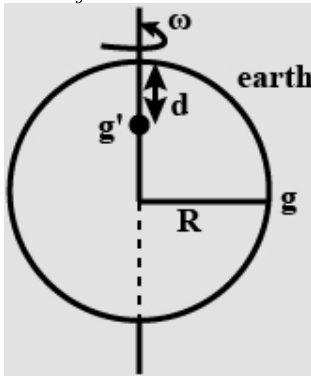
$g_{eff}$  due to rotation  $= g_{eff}$  due to depth

$$g - \omega^2 R = g \left(1 - \frac{d}{R}\right)$$

$$g - R\omega^2 = g - \frac{gd}{R}$$

$$-R\omega^2 = -\frac{gd}{R}$$

$$d = \frac{\omega^2 R^2}{g}$$



5. The gravitational force between two stones of mass  $1\text{ kg}$  each separated by a distance of  $1\text{ metre}$  in vacuum is

- A) Zero  
 B)  $6.675 \times 10^{-5}\text{ newton}$   
 C)  $6.675 \times 10^{-11}\text{ newton}$   
 D)  $6.675 \times 10^{-8}\text{ newton}$

**Solution : (Correct Answer: C )**

$$(c)F = G \frac{m_1 m_2}{r^2} = 6.675 \times \frac{1 \times 1}{1^2} \times 10^{-11} = 6.675 \times 10^{-11} N$$

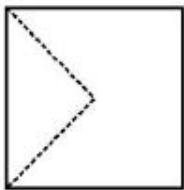
6. Two particle of masses  $1\text{ kg}$  and  $3\text{ kg}$  have position vector  $2\hat{i} + 3\hat{j} + 4\hat{k}$  and  $-2\hat{i} + 3\hat{j} - 4\hat{k}$  respectively. The centre of mass has a position vector

- A)  $-\hat{i} - 3\hat{j} - 2\hat{k}$   
 B)  $\hat{i} + 3\hat{j} - 2\hat{k}$   
 C)  $-\hat{i} + 3\hat{j} + 2\hat{k}$   
 D)  $-\hat{i} + 3\hat{j} - 2\hat{k}$

**Solution : (Correct Answer: D )**

$$\begin{aligned} \vec{r}_{cm} &= \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \\ &= \frac{1[2\hat{i} + 3\hat{j} + 4\hat{k}] + 3[-2\hat{i} + 3\hat{j} - 4\hat{k}]}{4} \\ &= -\hat{i} + 3\hat{j} - 2\hat{k} \end{aligned}$$

7. From the homogeneous square plate we cut a triangle (Figure). Side of the square is  $a$  and. the apex of the triangle is at the center of the square. Distance from the center of the square to the center of mass of the remainder of the plate is



- A)  $a/5$   
 B)  $a/3$   
 C)  $a/6$   
 D)  $a/9$

**Solution : (Correct Answer: D )**

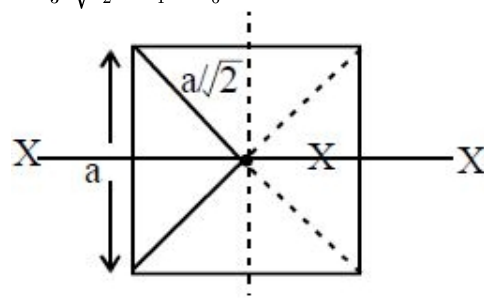
X – X in axis of symmetry so centre of mass lie on

X – X at centroid position of dotted triangle.

$\frac{2}{3}$  (median of triangle)

$$\frac{2}{3} \sqrt{\left(\frac{a}{\sqrt{2}}\right)^2 - \left(\frac{a}{2}\right)^2}$$

$$= \frac{2a}{3} \sqrt{\frac{1}{2} - \frac{1}{4}} = \frac{a}{6}$$



8. Consider the following statements

Assertion (A) : A cyclist always bends inwards while negotiating a curve

Reason (R) : By bending he lowers his centre of gravity Of these statements,

- A) both A and R are true and R is the correct explanation of A  
 B) both A and R are true but R is not the correct explanation of A  
 C) A is true but R is false  
 D) A is false but R is true

**Solution : (Correct Answer: B )**

9. Two masses  $m_1$  and  $m_2$  ( $m_1 > m_2$ ) are connected by massless flexible and inextensible string passed over massless and frictionless pulley. The acceleration of center of mass is

- A)  $\left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 g$   
 B)  $\frac{m_1 - m_2}{m_1 + m_2} g$   
 C)  $\frac{m_1 + m_2}{m_1 - m_2} g$   
 D) Zero

**Solution : (Correct Answer: A )**

$$(a) \text{ Acceleration of each mass} = a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) g$$

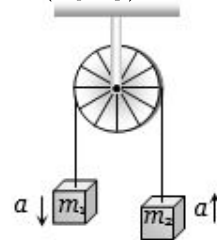
Now acceleration of centre of mass of the system

$$A_{cm} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2}{m_1 + m_2}$$

As both masses move with same acceleration but in opposite direction so

$$\vec{a}_1 = -\vec{a}_2 = a \text{ (let)}$$

$$\begin{aligned} \therefore A_{cm} &= \frac{m_1 a - m_2 a}{m_1 + m_2} \\ &= \left(\frac{m_1 - m_2}{m_1 + m_2}\right) \times \left(\frac{m_1 - m_2}{m_1 + m_2}\right) \times g \\ &= \left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 \times g \end{aligned}$$



10. Where will be the centre of mass on combining two masses  $m$  and  $M$  ( $M > m$ )

- A) Towards  $m$   
 B) Towards  $M$   
 C) Between  $m$  and  $M$   
 D) Anywhere

**Solution : (Correct Answer: B )**

Centre of mass is nearer to heavier mass

11. Identify the correct statements from the following:

(A) Work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket is negative.  
 (B) Work done by gravitational force in lifting a bucket out of a well by a rope tied to the bucket is negative.  
 (C) Work done by friction on a body sliding down an inclined plane is positive.  
 (D) Work done by an applied force on a body moving on a rough horizontal plane with uniform velocity is zero.  
 (E) Work done by the air resistance on an oscillating pendulum is negative.  
 Choose the correct answer from the options given below:

- A) B and E only  
 B) A and C only  
 C) B, D and E only  
 D) B and D only

**Solution : (Correct Answer: A )**

12. A particle of mass  $m$  is projected at an angle  $\alpha$  to the horizontal with initial velocity  $u$  m/s then work done by gravity during the time it reaches its highest point

- A)  $u^2 \sin^2 \alpha$   
 B)  $\frac{mu^2 \cos^2 \alpha}{2}$   
 C)  $-\frac{mu^2 \sin^2 \alpha}{2}$   
 D)  $\frac{mu^2 g \sin^2 \alpha}{2}$

**Solution : (Correct Answer: C )**

$$\vec{F} = mg(-\hat{j})$$

$$\vec{S} = \frac{R}{2}\hat{i} + H\hat{j}$$

$$\text{Work done by gravity} = \vec{F} \cdot \vec{S}$$

$$= -mgH = -mg \left( \frac{u^2 \sin^2 \alpha}{2g} \right)$$

13. A horizontal force of 5 N is required to maintain a velocity of 2 m/s for a block of 10 kg mass sliding over a rough surface. The work done by this force in one minute is....J

- A) 600      B) 60      C) 6      D) 6000

**Solution : (Correct Answer: A )**

$$(a) W = F \times s = F \times v \times t = 5 \times 2 \times 60 = 600 \text{ J}$$

14. A 60 kg weight is dragged on a horizontal surface by a rope upto 2 metres. If coefficient of friction is  $\mu = 0.5$ , the angle of rope with the surface is  $60^\circ$  and  $g = 9.8 \text{ m/sec}^2$ , then work done is ..... joules

- A) 294      B) 315      C) 588      D) 197

**Solution : (Correct Answer: B )**

(b) Let body is dragged with force  $P$ , making an angle  $60^\circ$  with the horizontal.

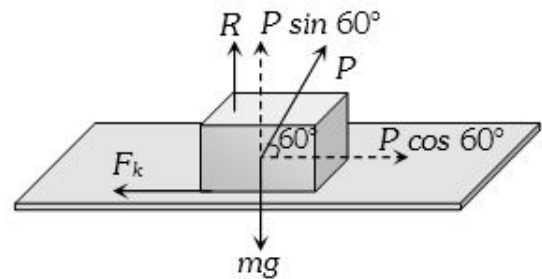
$$F_k = \text{Kinetic friction in the motion} = \mu_k R$$

$$\text{From the figure } F_k = P \cos 60^\circ \text{ and } R = mg - P \sin 60^\circ$$

$$\therefore P \cos 60^\circ = \mu_k (mg - P \sin 60^\circ)$$

$$\Rightarrow \frac{P}{2} = 0.5 \left( 60 \times 10 - \frac{P\sqrt{3}}{2} \right) \Rightarrow P = 315.1 \text{ N}$$

$$\therefore F_k = P \cos 60^\circ = \frac{315.1}{2} \text{ N Work done} = F_k \times s = \frac{315.1}{2} \times 2 = 315 \text{ Joule}$$



15. A force  $\vec{F} = 3\hat{i} + c\hat{j} + 2\hat{k}$  acting on a particle causes a displacement  $\vec{S} = -4\hat{i} + 2\hat{j} - 3\hat{k}$  in its own direction. If the work done is 6J, then the value of  $c$  will be

- A) 12      B) 6      C) 1      D) 0

**Solution : (Correct Answer: A )**

$$(a) W = \vec{F} \cdot \vec{S} = (3\hat{i} + c\hat{j} + 2\hat{k}) \cdot (-4\hat{i} + 2\hat{j} - 3\hat{k}) = -12 + 2c - 6$$

$$\text{Work done} = 6J \text{ (given)}$$

$$\therefore -12 + 2c - 6 = 6 \Rightarrow c = 12$$

16. A uniform rope lies on a horizontal table so that a part of it hangs over the edge. The rope begins to slide down when the length of the hanging part is 25% of the entire length. The coefficient of friction between the rope and the table is

- A) 0.25      B) 0.75      C) 0.33      D) 0.67

**Solution : (Correct Answer: C )**

$$\frac{3}{4} \mu_3 mg = \frac{1}{4} mg$$

$$\therefore \mu_s = \frac{1}{3} = 0.33$$

17. A block of mass  $M$  is being pulled along rough horizontal surface. The coefficient of friction between the block and the surface is  $\mu$ . If another block of mass  $M/2$  is placed on the block and it is again pulled on the surface, the coefficient of friction between the block and the surface will be

- A)  $\mu$       B)  $3\mu/2$   
 C)  $2\mu$       D)  $5\mu/2$

**Solution : (Correct Answer: A )**

Coefficient of friction is independent of the normal reaction. Hence, it will remain same.

18. A block of mass  $m$  is placed on a surface having vertical cross section given by  $y = x^2/4$ . If coefficient of friction is 0.5, the maximum height above the ground at which block can be placed without slipping is:

- A)  $1/4 \text{ m}$       B)  $1/2 \text{ m}$

- C)  $1/6$  m                      D)  $1/3$  m

**Solution : (Correct Answer: A)**

$$\frac{dy}{dx} = \tan \theta = \frac{x}{2} = \mu = \frac{1}{2}$$

$$x = 1, y = 1/4$$

19. A block of mass  $m$  rests on a rough inclined plane. The coefficient of friction between the surface and the block is  $\mu$ . At what angle of inclination  $\theta$  of the plane to the horizontal will the block just start to slide down the plane?

- A)  $\theta = \tan^{-1} \mu$                       B)  $\theta = \cos^{-1} \mu$   
C)  $\theta = \sin^{-1} \mu$                       D)  $\theta = \sec^{-1} \mu$

**Solution : (Correct Answer: A)**

The various forces acting on the block are as shown in the figure.

From figure

$$mg \sin \theta = f \quad \dots (i)$$

$$mg \cos \theta = N \quad \dots (ii)$$

Divide (i) by (ii), we get

$$\tan \theta = \frac{f}{N} = \frac{\mu N}{N}$$

$$\text{or } \theta = \tan^{-1}(\mu)$$

20. A car travels north with a uniform velocity. It goes over a piece of mud which sticks to the tyre. The particles of the mud, as it leaves the ground are thrown

- A) Vertically upwards      B) Vertically inwards  
C) Towards north              D) Towards south

**Solution : (Correct Answer: D)**

(d) The particle performing circular motion flies off tangentially.

21. An athlete does not come to rest immediately after crossing the winning line due to the

- A) Inertia of rest                      B) Inertia of motion  
C) Inertia of direction              D) None of these

**Solution : (Correct Answer: B)**

(b)

While running athlete is in the state of motion. So due to inertia of motion athlete does not come to rest.

22. *Assertion* : Use of ball bearings between two moving parts of a machine is a common practice.  
*Reason* : Ball bearings reduce vibrations and provide good stability.

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.  
B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

- C) If the Assertion is correct but Reason is incorrect.

- D) If both the Assertion and Reason are incorrect.

**Solution : (Correct Answer: C)**

Ball bearing are used to convert sliding friction to rolling friction. Sliding friction is less than rolling friction

23. Newton's first law of motion describes the following

- A) Energy                                      B) Work  
C) Inertia                                      D) Moment of inertia

**Solution : (Correct Answer: C)**

(c) Newton's first law of motion defines the inertia of body. It states that every body has a tendency to remain in its state (either rest or motion) due to its inertia.

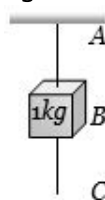
24. When a train stops suddenly, passengers in the running train feel an instant jerk in the forward direction because

- A) The back of seat suddenly pushes the passengers forward  
B) Inertia of rest stops the train and takes the body forward  
C) Upper part of the body continues to be in the state of motion whereas the lower part of the body in contact with seat remains at rest  
D) Nothing can be said due to insufficient data

**Solution : (Correct Answer: C)**

This is due to inertia. When the speeding bus stops suddenly, lower part of the body comes to rest while the upper part of the body tends to maintain uniform motion. Hence, the passenger's are thrown forward.

25. A mass of  $1\text{ kg}$  is suspended by a string  $A$ . Another string  $C$  is connected to its lower end (see figure)., if the string  $C$  is stretched slowly, then



- A) The portion  $AB$  of the string will break  
B) The portion  $BC$  of the string will break  
C) None of the strings will break  
D) None of the above

**Solution : (Correct Answer: A)**

(a) When the spring  $C$  is stretched slowly, the tension in  $A$  is greater than that of  $C$ , because of the weight  $mg$  and the former reaches breaking point earlier.

26. Velocity of a particle moving in a curvilinear path in a horizontal  $XY$  plane varies with time as  $\vec{v} = (2t\hat{i} + t^2\hat{j})$  m/s. Here,  $t$  is in second. At  $t = 1$  s

- A) acceleration of particle is  $8 \text{ m/s}^2$   
 B) tangential acceleration of particle is  $\frac{4}{\sqrt{5}} \text{ m/s}^2$   
 C) radial acceleration of particle is  $\frac{6}{\sqrt{5}} \text{ m/s}^2$   
 D) radius of curvature to the path is  $\frac{5\sqrt{5}}{2} \text{ m}$

**Solution : (Correct Answer: D)**

$$\vec{V} = 2t\hat{i} + t^2\hat{j}$$

$$\vec{a} = 2\hat{i} + 2t\hat{j}$$

$$|\vec{V}| = V = \sqrt{4t^2 + t^4}$$

$$a_t = \left(\frac{dV}{dt}\right) = \frac{1}{2\sqrt{4t^2 + t^4}} \times (8t + 4t^3)$$

$$\text{at } t = 1 \quad a_t = \left(\frac{6}{\sqrt{5}}\right) \text{ m/s}^2$$

$$a = \sqrt{2^2 + (2t)^2} = \sqrt{8} \text{ m/s}^2$$

$$r = \sqrt{5}$$

$$a_c = \sqrt{a^2 - a_t^2} = \sqrt{8 - \frac{36}{5}} = \left(\frac{2}{\sqrt{5}}\right)$$

$$R = \left(\frac{V^2}{a_c}\right) = \left(\frac{5}{\frac{2}{\sqrt{5}}}\right) = \frac{5\sqrt{5}}{2}$$

$$R = \left(\frac{5\sqrt{5}}{2}\right) \text{ m}$$

27. A particle moves in the  $xy$  plane with a constant acceleration ' $g$ ' in the negative  $y$ -direction. Its equation of motion is  $y = ax - bx^2$ , where  $a$  and  $b$  are constants. Which of the following are correct?

- A) The  $x$ -component of its velocity is constant.  
 B) At the origin, the  $y$ -component of its velocity is  $a\sqrt{\frac{g}{2b}}$ .  
 C) At the origin, its velocity makes an angle  $\tan^{-1}(a)$  with the  $x$ -axis.  
 D) All of the above

**Solution : (Correct Answer: D)**

As in this it is given that the particle will only have the acceleration in the negative  $y$ -direction so the acceleration in the  $x$ -direction is zero so  $a_x = 0$

so the velocity in the  $x$ -direction will be constant as acceleration in the  $x$ -direction is

$$\text{zero now } y = ax - bx^2$$

so now differentiate the equation so we get

$$\frac{dy}{dt} = a \frac{dx}{dt} - 2bx \frac{dx}{dt}$$

$$\text{now differentiate this again } \frac{d^2y}{dt^2} = a \frac{d^2x}{dt^2}$$

$$-2b \left( \left(\frac{dx}{dt}\right)^2 + x \frac{d^2x}{dt^2} \right)$$

$$\text{that is } a_y = aa_x - 2b(v_x^2 + 2xa_x)$$

$$\text{as } a_x = 0 \text{ and } a_y = -g$$

$$\text{so } -g = -2bv_x^2$$

$$\text{so } v_x = \sqrt{g/2b}$$

$$\text{so as we know } v_y = av_x - 2bxv_x$$

$$\text{as at origin means } x = 0 \quad v_y = av_x$$

$$\text{so } v_y = a\sqrt{g/2b}$$

28. A particle moves in a circle of radius  $25 \text{ cm}$  at two revolutions per sec. The acceleration of the particle in  $\text{m/s}^2$  is

- A)  $\pi^2$       B)  $8\pi^2$       C)  $4\pi^2$       D)  $2\pi^2$

**Solution : (Correct Answer: C)**

$$r = 25 \times 10^{-2} \text{ m}, f = 2/\text{sec}$$

$$\therefore \omega = 2\pi f = 4\pi \text{ rad/sec}$$

$$\text{Acceleration} = \omega^2 r = (4\pi)^2 \times 25 \times 10^{-2}$$

$$= 16 \times 25 \times 10^{-2} \pi^2 \text{ m/s}^2 = 4\pi^2 \text{ m/s}^2$$

29. A particle is projected vertically upwards from  $O$  with velocity  $v$  and a second particle is projected at the same instant from  $P$  (at a height  $h$  above  $O$ ) with velocity  $v$  at an angle of projection  $\theta$ . The time when the distance between them is minimum is

- A)  $\frac{h}{2v \sin \theta}$       B)  $\frac{h}{2v \cos \theta}$   
 C)  $h/v$       D)  $h/2v$

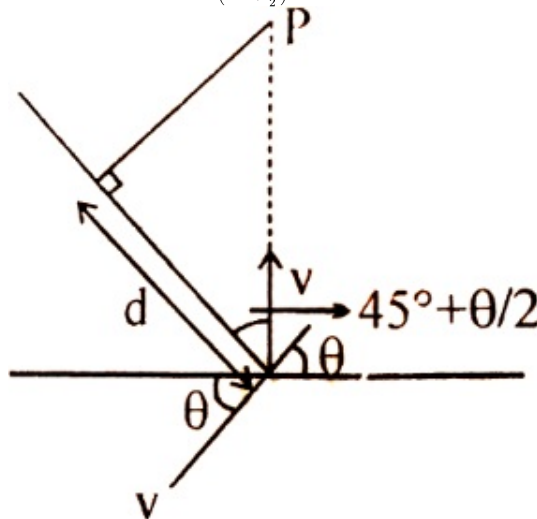
**Solution : (Correct Answer: D)**

In from of ball thrown from  $p$

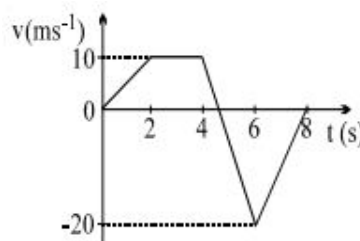
$$v_{\text{resultant}} = 2v \cos \left( \frac{90^\circ + \theta}{2} \right)$$

$$= 2v \cos (45^\circ + \theta/2)$$

$$t = \frac{d}{v_{\text{resultant}}} = \frac{h \cos(45^\circ + \frac{\theta}{2})}{2v \cos(45^\circ + \frac{\theta}{2})} = \frac{h}{2v}$$



30. The figure shows a velocity-time graph of a particle moving along a straight line. The total distance travelled by the particle is .....  $m$



- A) 66.6      B) 51.6      C) 0      D) 36.6

**Solution : (Correct Answer: A)**

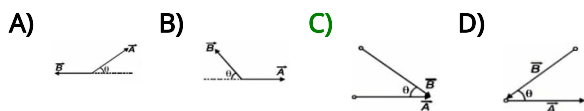
Distance = Area covered by  $v-t$  graph on time

$$\text{axis} \Rightarrow \frac{1}{2} \left( +\frac{14}{3} \right) \times 10 + \frac{1}{2} \times \left( 8 - \frac{14}{3} \right) \times 20$$

$$= \frac{100}{3} + \frac{100}{3} = \frac{200}{3} \Rightarrow 66.6 \text{ m}$$



31. Let  $\theta$  be the angle between vectors  $\vec{A}$  and  $\vec{B}$ . Which of the following figures correctly represents the angle  $\theta$ ?



**Solution : (Correct Answer: C)**

(c)

To find angle between vectors, they will be joined either head to head or tail to tail.

32. If  $|\vec{v}_1 + \vec{v}_2| = |\vec{v}_1 - \vec{v}_2|$  and  $\vec{v}_1$  and  $\vec{v}_2$  are finite, then

- A)  $\vec{v}_1$  is parallel to  $\vec{v}_2$   
 B)  $\vec{v}_1 = \vec{v}_2$   
 C)  $|\vec{v}_1| = |\vec{v}_2|$   
 D)  $\vec{v}_1$  and  $\vec{v}_2$  are mutually perpendicular

**Solution : (Correct Answer: D)**

33. The position vectors of points  $A, B, C$  and  $D$  are  $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ ,  $\vec{B} = 4\hat{i} + 5\hat{j} + 6\hat{k}$ ,  $\vec{C} = 7\hat{i} + 9\hat{j} + 3\hat{k}$  and  $\vec{D} = 4\hat{i} + 6\hat{j}$  then the displacement vectors  $\vec{AB}$  and  $\vec{CD}$  are

- A) Perpendicular  
 B) Parallel  
 C) Antiparallel  
 D) Inclined at an angle of  $60^\circ$

**Solution : (Correct Answer: C)**

$$\vec{AB} = \vec{B} - \vec{A} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{CD} = \vec{D} - \vec{A} = -3\hat{i} - 3\hat{j} - 3\hat{k}$$

$$\Rightarrow \vec{CD} = -3\vec{AB}$$

34.  $0.4\hat{i} + 0.8\hat{j} + c\hat{k}$  represents a unit vector when  $c$  is

- A)  $-0.2$  B)  $\sqrt{0.2}$  C)  $\sqrt{0.8}$  D) 0

**Solution : (Correct Answer: B)**

$$(b) \sqrt{(0.4)^2 + (0.8)^2 + c^2} = 1$$

$$\Rightarrow 0.16 + 0.64 + c^2 = 1 \Rightarrow c = \sqrt{0.2}$$

35. Two adjacent sides of a parallelogram are represented by the two vectors  $\hat{i} + 2\hat{j} + 3\hat{k}$  and  $3\hat{i} - 2\hat{j} + \hat{k}$ . What is the area of parallelogram

- A) 8 B)  $8\sqrt{3}$  C)  $3\sqrt{8}$  D) 192

**Solution : (Correct Answer: B)**

$$(b) \text{Area of parallelogram} = |\vec{A} \times \vec{B}|$$

$$= (\hat{i} + 2\hat{j} + 3\hat{k}) \times (3\hat{i} - 2\hat{j} + \hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 3 & -2 & 1 \end{vmatrix}$$

$$= (8)\hat{i} + (8)\hat{j} - (8)\hat{k}$$

$$\text{Magnitude} = \sqrt{64 + 64 + 64} = 8\sqrt{3}$$

36. A particle is moving along a circle such that it completes one revolution in 40 seconds. In 2 minutes 20 seconds, the ratio  $\frac{|\text{displacement}|}{\text{distance}}$  is .....

- A) 0 B)  $\frac{1}{7}$  C)  $\frac{2}{7}$  D)  $\frac{1}{11}$

**Solution : (Correct Answer: D)**

(d)

$$T = 40 \text{ s}$$

If  $t = 2 \text{ minute } 20 \text{ second}$

$$= 2 \times 60 + 20$$

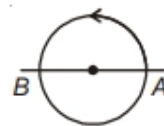
$$= 140 \text{ s}$$

So, it has completed  $3\frac{1}{2}$  revolution.

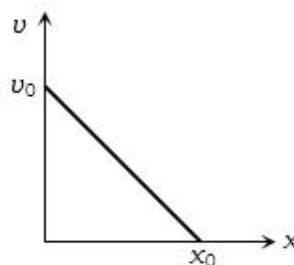
$$\text{Distance travelled} = 3 \times 2\pi R + \pi R = 7\pi R$$

$$\text{Displacement} = 2R$$

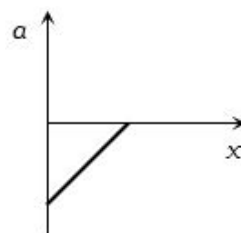
$$\frac{|\text{Displacement}|}{\text{Distance}} = \frac{2R}{7\pi R} = \frac{2}{7 \times \frac{22}{7}} = \frac{1}{11}$$



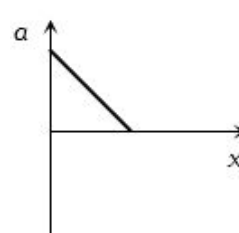
37. The given graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement



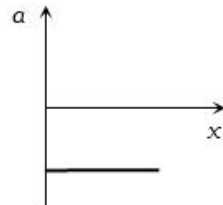
A)



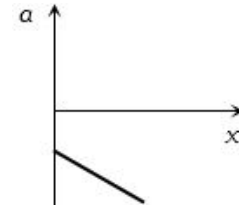
B)



C)



D)



**Solution : (Correct Answer: A)**

(a) Given line have positive intercept but negative slope. So its equation can be written as

$$v = -mx + v_0 \dots (i) \quad [\text{where } m = \tan \theta = \frac{v_0}{x_0}]$$

By differentiating with respect to time we get  $\frac{dv}{dt}$   
 $= -m \frac{dx}{dt} = -mv$

Now substituting the value of  $v$  from eq. (i) we get

$$\frac{dv}{dt} = -m[-mx + v_0] = m^2x - mv_0$$

$$\therefore a = m^2x - mv_0$$

i.e. the graph between  $a$  and  $x$  should have positive slope but negative intercept on  $a$ -axis. So graph (a) is correct.

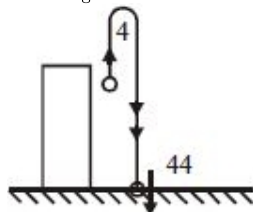
38. A ball is thrown vertically upwards from the top of a tower with a velocity  $u$ . This ball reaches the ground level with a velocity  $4u$ . The height of the tower is

- A)  $\frac{3u^2}{g}$  B)  $\frac{4u^2}{g}$   
 C)  $\frac{6u^2}{g}$  D)  $\frac{15u^2}{2g}$

**Solution : (Correct Answer: D)**

$$(4u)^2 = u^2 + 2g \cdot h$$

$$h = \frac{15u^2}{2g}$$



39. Speed of two identical cars are  $u$  and  $4u$  at a specific instant. The ratio of the respective distances in which the two cars are stopped from that instant is

- A) 1 : 1 B) 1 : 4 C) 1 : 8 D) 1 : 16

**Solution : (Correct Answer: D)**

$$(d) S \propto u^2 \Rightarrow \frac{S_1}{S_2} = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

40. Which of the following is a one dimensional motion

- A) Landing of an aircraft  
 B) Earth revolving around the sun  
 C) Motion of wheels of a moving train  
 D) Train running on a straight track

**Solution : (Correct Answer: D)**

Motion of train on straight track is a one dimensional motion whereas everything else is a two dimensional or three dimensional activity.

41. A quantity  $f$  is given by  $f = \sqrt{\frac{hc^5}{G}}$  where  $c$  is speed of light,  $G$  universal gravitational constant and  $h$  is the Planck's constant. Dimension of  $f$  is that of

- A) Momentum B) Area  
 C) Energy D) Volume

**Solution : (Correct Answer: C)**

$$[h] = M^1L^2T^{-1}$$

$$[C] = L^1T^{-1}$$

$$[G] = M^{-1}L^3T^{-2}$$

$$[f] = \sqrt{\frac{M^1L^2T^{-1} \times L^5T^{-5}}{M^{-1}L^3T^{-2}}} = M^1L^2T^{-2}$$

42. The relative error in the determination of the surface area of a sphere is  $\alpha$ . Then the relative error in the determination of its volume is

- A)  $\frac{2}{3}\alpha$  B)  $\frac{5}{2}\alpha$   
 C)  $\frac{3}{2}\alpha$  D)  $\alpha$

**Solution : (Correct Answer: C)**

Relative error in surface area,

$$\frac{\Delta s}{s} = 2 \times \frac{\Delta r}{r} = \alpha \text{ and relative error in}$$

$$\text{volume, } \frac{\Delta v}{v} = 3 \times \frac{\Delta r}{r}$$

$\therefore$  Relative error in volume w.r.t relative error in area,

$$\frac{\Delta v}{v} = \frac{3}{2}\alpha$$

43. Which of the following is the unit of specific heat

- A)  $J/kg^\circ C$  B)  $J/kg^\circ C^{-1}$   
 C)  $kg^\circ C/J$  D)  $J/kg^\circ C^{-2}$

**Solution : (Correct Answer: A)**

$$(a) c = \frac{Q}{m \cdot \Delta \theta} \rightarrow \frac{J}{kg \times ^\circ C}$$

44. If  $V$  denotes the potential difference across the plates of a capacitor of capacitance  $C$ , the dimensions of  $CV^2$  are

- A) Not expressible in  $MLT$   
 B)  $MLT^{-2}$   
 C)  $M^2LT^{-1}$   
 D)  $ML^2T^{-2}$

**Solution : (Correct Answer: D)**

$$(d) \frac{1}{2} CV^2 = \text{Stored energy in a capacitor} = [ML^2T^{-2}]$$

45. Which is not a unit of electric field

- A)  $NC^{-1}$  B)  $Vm^{-1}$   
 C)  $JC^{-1}$  D)  $JC^{-1}m^{-1}$

**Solution : (Correct Answer: C)**

SI unit of electric field strength is newtons per coulomb ( $N/C$ ) or volts per meter ( $V/m$ ).

It is also expressed as  $JC^{-1}m^{-1}$ .

## Chemistry

46. Calculate the enthalpy change when 50 ml of 0.01 M  $Ca(OH)_2$  reacts with 25 ml of 0.01 M  $HCl$ . Given that  $\Delta H^\circ$  for neutralisation of a strong acid and a strong base is  $140 \text{ kJ mol}^{-1}$  ..... kJ

- A) 14 B) 35 C) 10 D) 7.5

**Solution : (Correct Answer: B)**

$$\text{Number of moles of } HCl = \frac{MV}{1000} = \frac{0.01 \times 25}{1000}$$

$$= 25 \times 10^{-5}$$

$$n_{H^+} = 25 \times 10^{-5}$$

$$\text{Number of moles of } Ca(OH)_2$$

$$= \frac{MV}{1000} = \frac{0.01 \times 50}{1000} = 50 \times 10^{-5}$$

$$n_{OH^-} = 2 \times 50 \times 10^{-5} = 10^{-3}$$



In the process of neutralisation  $25 \times 10^{-5}$  mole

$H^+$ ,  $OH^-$  will be neutralised

$$\Delta H = 140 \times 25 \times 10^{-5} \text{ kcal} = 3500 \times 10^{-5} \text{ kcal}$$

$$= 35 \text{ cal}$$

47. An ideal gas undergoes isothermal expansion at constant pressure. During the process

- A) enthalpy increases but entropy decreases  
**B) enthalpy remains constant but entropy increases**  
 C) enthalpy decreases but entropy increases.  
 D) Both enthalpy and entropy remain constant.

**Solution : (Correct Answer: B)**

$$\Delta H = nC_p \Delta T = 0 \text{ means } \Delta H \text{ constant.}$$

$$\Delta S = nR \ln \left( \frac{v_f}{v_i} \right) \geq 0 \Delta S \text{ increases.}$$

48. The heat of hydrogenation for 3-methylbutene and 2-pentene are  $-30 \text{ kcal/mol}$  and  $-28 \text{ kcal/mol}$  respectively. The heats of combustion of 2-methylbutane and pentane are  $-784 \text{ kcal/mol}$  and  $-782 \text{ kcal/mol}$  respectively. All the values are given under standard conditions. Taking into account that combustion of both alkanes give the same products, what is  $\Delta H$  (in  $\text{kcal/mol}$ ) for the following reaction under same conditions?



- A) 0      **B) -4**      C) -2      D) 2

**Solution : (Correct Answer: B)**

$$\Delta H_1 = -30 \text{ kcal/mol}, \Delta H_2 = (-784 + 782) \text{ kcal/mol},$$

$$\Delta H_3 = 28 \text{ kcal/mol}$$

$$\Rightarrow \Delta H_1 + \Delta H_2 + \Delta H_3 = -4 \text{ kcal/mol}$$



49. The equilibrium constant of a reaction at  $298 \text{ K}$  is  $5 \times 10^{-3}$  and at  $1000 \text{ K}$  is  $2 \times 10^{-5}$ . What is the sign of  $\Delta H$  for the reaction

- A)  $\Delta H = 0$   
**B)  $\Delta H$  is negative**  
 C)  $\Delta H$  is positive  
 D) None of these

**Solution : (Correct Answer: B)**

(b) Equilibrium constant decreases with temperature and  $\Delta H$  is also decreases so it is  $-ve$ .

50. The standard enthalpy of the decomposition of  $N_2O_4$  to  $NO_2$  is  $58.04 \text{ kJ}$  and standard entropy of this reaction is  $176.7 \text{ J/K}$ . The standard free energy change for this reaction at  $25^\circ \text{C}$  is  $\dots \text{kJ}$

- A) 539      B) -539      C) -5.39      **D) 5.39**

**Solution : (Correct Answer: D)**

The computation are as expressed below,

$$\Delta G = \Delta H - T\Delta S$$

$$= (58.04 \times 10^3 \text{ J}) - [(273 + 25) \text{ K} \times 176.7 \text{ J/K}]$$

$$= 5383.4 \text{ J} = 5.39 \text{ kJ}$$

51. The compound with negative heat of formation are known as

- A) Endothermic compound  
**B) Exothermic compound**  
 C) Heat of formation compound  
 D) None of the above

**Solution : (Correct Answer: B)**

(b)  $\Delta H = -ve$  for exothermic compound.

52. If a chemical reaction is accompanied by the evolution of heat, it is

- A) Catalytic      B) Photochemical  
 C) Endothermic      **D) Exothermic**

**Solution : (Correct Answer: D)**

(d) For exothermic reaction heat is evolved.

53. Heat of combustion of a substance

- A) Is always positive  
**B) Is always negative**  
 C) Is equal to heat of formation  
 D) Nothing can be said without reaction

**Solution : (Correct Answer: B)**

Heat of combustion is always negative.

54. For the isothermal expansion of an ideal gas

- A)  $E$  and  $H$  increases  
 B)  $E$  increases but  $H$  decreases  
 C)  $H$  increases but  $E$  decreases  
**D)  $E$  and  $H$  are unaltered**

**Solution : (Correct Answer: D)**

(d) In isothermal reversible process ideal gas has constant volume and so  $\Delta E = 0$  and  $\Delta H = \Delta E = 0$ .

55. Select equation having exothermic step

- A)  $S_{(g)}^- \rightarrow S_{(g)}^{-2}$   
**B)  $Na_{(g)}^+ + Cl_{(g)}^- \rightarrow NaCl_{(s)}$**   
 C)  $N_{(g)} \rightarrow N_{(g)}^-$   
 D)  $Al_{(g)}^{+2} \rightarrow Al_{(g)}^{+3}$

**Solution : (Correct Answer: B)**

$Na_{(g)}^+ + Cl_{(g)}^- \rightarrow NaCl_{(s)}$  lattice energy (exothermic)

56. Match list I with list II and select the correct answer

| list I (species) | list II (O - N - O angle) |
|------------------|---------------------------|
| (A) $NO_2^+$     | (1) $180^\circ$           |
| (B) $NO_2$       | (2) $132^\circ$           |
| (C) $NO_2^-$     | (3) $120^\circ$           |
| (D) $NO_3^-$     | (4) $115^\circ$           |
|                  | (5) $109^\circ$           |

- A) A - 5, B - 4, C - 3, D - 2

B)  $A - 5, B - 2, C - 4, D - 3$

C)  $A - 1, B - 2, C - 4, D - 3$

D)  $A - 1, B - 4, C - 3, D - 2$

**Solution : (Correct Answer: C)**

$NO_2^+$  is  $sp$  and linear i.e. bond angle is  $180^\circ$ .

$NO_2$  has a bond angle of  $132^\circ$

$NO_2^-$  is  $sp^2$  and bond angle is less than  $120^\circ$  due to lone pair bond pair repulsion.

$NO_3^-$  is  $sp^2$  and trigonal planar i.e. bond angle is  $120^\circ$ .

57. The compound with the highest degree of covalency is

A)  $NaCl$  B)  $MgCl_2$  C)  $AgCl$  D)  $CsCl$

**Solution : (Correct Answer: C)**

The lower the difference in electro negativities, more is degree of covalency.  $Ag$  is not as electro positive as alkali or alkaline earth metals.

58. The ionic bonds  $X^+Y^-$  are formed when  
(I) electron affinity of  $Y$  is high (II) ionization energy of  $X$  is low  
(III) lattice energy of  $XY$  is high (IV) lattice energy of  $XY$  is low  
Choose the correct code

A) I and II B) I and III  
C) I, II and III D) All

**Solution : (Correct Answer: C)**

59. The hydration energy of  $Mg^{2+}$  ions is lesser than that of

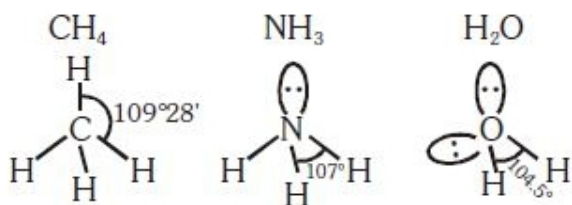
A)  $Al^{3+}$  B)  $Ba^{2+}$   
C)  $Na^+$  D) None of these

**Solution : (Correct Answer: A)**

60. Consider the molecules  $CH_4, NH_3$  and  $H_2O$ . Which of the given statements is false?

A) The  $H - O - H$  bond angle in  $H_2O$  is smaller than the  $H - N - H$  bond angle in  $NH_3$ .  
B) The  $H - C - H$  bond angle in  $CH_4$  is larger than the  $H - N - H$  bond angle in  $NH_3$ .  
C) The  $H - C - H$  bond angle in  $CH_4$ , the  $H - N - H$  bond angle in  $NH_3$ , and the  $H - O - H$  bond angle in  $H_2O$  are all greater than  $90^\circ$ .  
D) The  $H - O - H$  bond angle in  $H_2O$  is larger than the  $H - C - H$  bond angle in  $CH_4$ .

**Solution : (Correct Answer: D)**



61. In which of the following pairs molecules have bond order three and are isoelectronics

A)  $CN^-, CO$  B)  $NO^+, CO^+$   
C)  $CN^-, O_2^+$  D)  $CO, O_2^+$

**Solution : (Correct Answer: A)**

(a)  $CN^- = 14$  electrons ;  $CO = 14$  electrons

$$B.O. = \frac{1}{2}[10 - 4] = \frac{6}{2} = 3.$$

62. The smallest bond angle is found in

A)  $IF_7$  B)  $CH_4$   
C)  $BeF_2$  D)  $BF_3$

**Solution : (Correct Answer: A)**

(a) It shows  $sp^3d^3$ -hybridization. Hence the bond angle is about  $72^\circ$ .

63. When  $NaCl$  is dissolved in water the sodium ion becomes

A) Oxidized B) Reduced  
C) Hydrolysed D) Hydrated

**Solution : (Correct Answer: D)**

(d) When sodium chloride is dissolved in water, the sodium ion is hydrated.

64. The law of triads is applicable to a group of

A)  $Cl, Br, I$  B)  $C, N, O$   
C)  $Na, K, Rb$  D)  $H, O, N$

**Solution : (Correct Answer: A)**

According to the law of triads the atomic wt of the middle element is arithmetic mean of I and III.

$$\text{At wt of } Br = \frac{\text{At. wt of } Cl + \text{At wt of } I}{2}$$

65. The electron affinity of  $Be$  is similar to :-

A)  $He$  B)  $B$  C)  $Li$  D)  $Na$

**Solution : (Correct Answer: A)**

$He$  being noble gas has complete 1st shell. It has completed its duplet. While  $Be$  has electronic configuration  $1s^2, 2s^2$  so  $Be$  also has completely filled  $2s$  subshell so we can say that, The electron affinity of  $Be$  is similar to  $Be$  among the given options.

66. Which will show maximum non-metallic character

A)  $B$  B)  $Be$  C)  $Mg$  D)  $Al$

**Solution : (Correct Answer: A)**

Non-metallic character decreases as we move down the group and across a period non-metallic character increases from left to right.

So here  $B$  will show maximum non-metallic character.

67. The order of the magnitude of first ionisation potentials of  $Be, B, N$  and  $O$  is

A)  $N > O > Be > B$   
B)  $N > Be > O > B$   
C)  $Be > B > N > O$

D)  $B > Be > O > N$

**Solution : (Correct Answer: A)**

(a)  $N > O > Be > B$  Ist ionisation energy of ( $n$ ) because of half filled  $p$ -orbital.

68. Which of the following ions has the smallest radius

A)  $Be^{2+}$     B)  $Li^+$     C)  $O^{2-}$     D)  $F^-$

**Solution : (Correct Answer: A)**

It's Obvious

69. The chemistry of lithium is very similar to that of magnesium even though they are placed in different groups

A) Both are found together in nature  
B) Both have nearly the same size  
C) Both have similar electronic configuration  
D) The ratio of their charge to size is nearly the same

**Solution : (Correct Answer: D)**

A Diagonal Relationship is said to exist between certain pairs of diagonally adjacent elements in the second and third periods of the periodic table. These pairs ( $Li$  and  $Mg$ ,  $Be$  and  $Al$ ,  $B$  and  $Si$  etc.) exhibit similar properties; for example, Boron and Silicon are both semiconductors, form halides that are hydrolyzed in water and have acidic oxides. Such a relationship occurs because crossing and descending the periodic table have opposing effects. On crossing a period of the periodic table, the size of the atoms decreases, and on descending a group the size of the atoms increases. Similarly, on moving along the period the elements become progressively more covalent, less reducing and more electronegative, whereas on descending the group the elements become more ionic, more basic and less electronegative. Thus, on both descending a group and crossing by one element the changes cancel each other out, and elements with similar properties which have similar chemistry are often found - the atomic size, electronegativity, properties of compounds (and so forth) of the diagonal members are similar.

The chemistry of lithium is very similar to that of magnesium even though they are placed in different groups. Its reason is that the ratio of their charge to size is nearly the same

70. As one moves along a given row in the periodic table, ionization energy

A) Remains same  
B) Increases from left to right  
C) First increases, then decreases  
D) Decreases from left to right

**Solution : (Correct Answer: B)**

(b) Increases from left to right. Since, the size decreases.

71. Who is called the father of chemistry

A) Faraday    B) Priestley  
C) Rutherford    D) Lavoisier

**Solution : (Correct Answer: D)**

Antoine Lavoisier is called the father of modern chemistry. He recognized and named oxygen and isolated the major components of air.

72. The most important active step in the development of periodic table was taken by

A) Mendeleev    B) Dalton  
C) Avogadro    D) Cavendish

**Solution : (Correct Answer: A)**

The most important active step in the development of periodic table was taken by Mendeleev since though others have also made significant contributions towards classification of elements, he elevated the progress made in that direction much more than anyone else.

73. If  $n = 2$  for  $He^+$  ion than ..... $\overset{\circ}{A}$  out the wave length

A) 3.33    B) 6.42    C) 1.47    D) 2.37

**Solution : (Correct Answer: A)**

As  $n = 2$  and  $Z = 2$  for  $He^+$  ion, so the wave length is calculated below.

$$2\pi r = n\lambda \quad \therefore r = 0.529 \frac{n^2}{Z}$$

$$2\pi \times 0.529 \frac{n^2}{Z} A = n\lambda$$

$$\lambda = \frac{2 \times 3.14 \times 0.529 \times 2}{2} A$$

$$\lambda = 3.33 \overset{\circ}{A}$$

74. The energy of an electron in excited hydrogen atom is  $-3.4 \text{ eV}$ . Then, according to Bohr's theory the angular momentum of this electron in  $Jsec$  is

A)  $2.1 \times 10^{-34}$   
B)  $4.2 \times 10^{-34}$   
C)  $6.6 \times 10^{-34}$   
D)  $3.2 \times 10^{-33}$

**Solution : (Correct Answer: A)**

$$E_n \text{ for H atom} = \frac{E_1}{n^2}$$

$$E_n = \frac{-13.6}{n^2} \Rightarrow n^2 = \frac{-13.6}{E_n}$$

$$\Rightarrow n^2 = \frac{-13.6}{-3.4} = 4$$

$$\Rightarrow n = 2$$

$$\Rightarrow \text{angular momentum} = \frac{nh}{2\pi} = \frac{2 \times h}{2\pi} = \frac{h}{\pi}$$

$$= \frac{6.6 \times 10^{-34}}{3.14}$$

$$= 2.1 \times 10^{-34} Jsec$$

75. The ( $e/m$ ) for positive rays in comparison to cathode rays is

A) Very low    B) Very high  
C) Same    D) None

**Solution : (Correct Answer: A )**

Cathode rays contains electrons and positive rays contain positive charged gaseous atom mass of electron is negligible in comparison to positive charge atom

$\therefore e/m$  ratio for cathode rays is very low.

76. Select the correct statements for quantum numbers.

- (i) Magnetic quantum no. ( $m_l$ ) gives information about the spatial orientation of orbitals with respect to standard set of co-ordinate axis.  
(ii) Electron spin quantum no. is represented by ' $s$ ' and have value ' $\frac{1}{2}$ '  
(iii) Principal quantum no. ( $n$ ) determine the size of the orbitals and also to a large extent of the energy of the orbitals.

- A) Only (i), (iii)      B) Only (iii)  
C) Only (i)      D) (i), (ii), (iii)

**Solution : (Correct Answer: A )**

77. The total number of electrons that can be accommodated in all the orbitals having principal quantum number 2 and azimuthal quantum number 1 is

- A) 2      B) 4      C) 6      D) 8

**Solution : (Correct Answer: C )**

The total number of electrons that can be accommodated in all the orbitals having principal quantum number 2 and the azimuthal quantum number 1 is 6 It is a p orbital with  $p_x, p_y, p_z$  sub orbitals each of which can accommodate 2 electrons each.

78. For a given value of quantum number  $l$ , the number of allowed values of  $m$  is given by

- A)  $l + 2$       B)  $2l + 2$   
C)  $2l + 1$       D)  $l + 1$

**Solution : (Correct Answer: C )**

(c)  $m = -l$  to  $+l$  including zero.

79. The ratio of specific charge of a proton and an  $\alpha$ -particle is

- A) 2 : 1      B) 1 : 2      C) 1 : 4      D) 1 : 1

**Solution : (Correct Answer: A )**

(a) Charge on proton = +1 unit, charge on  $\alpha$  particle = +2 units, 2 : 1.

80. What is the ratio of mass of an electron to the mass of a proton

- A) 1 : 2      B) 1 : 1      C) 1 : 1837      D) 1 : 3

**Solution : (Correct Answer: C )**

(c) Electron  $\frac{1}{1837}$  time lighter than proton so their mass ratio will be 1 : 1837

81. Heaviest particle is

- A) Meson    B) Neutron    C) Proton    D) Electron

**Solution : (Correct Answer: B )**

(b) Mass of neutron is greater than that of proton, meson and electron.

Mass of neutron = mass of proton + mass of electron

82. The mole fraction of urea in an aqueous urea solution containing 900g of water is 0.05 . If the density of the solution is  $1.2 \text{ g cm}^{-3}$ , the molarity of urea solution is. . . . (Given data : Molar masses of urea and water are  $60 \text{ g mol}^{-1}$  and  $18 \text{ g mol}^{-1}$ , respectively)

- A) 2.50      B) 2.55      C) 2.60      D) 2.98

**Solution : (Correct Answer: D )**

$$X_{\text{urea}} = 0.05 = \frac{n}{n+50}$$

$$19n = 50$$

$$n = 2.6315$$

$$V_{\text{sol}} = \frac{(2.6315 \times 60 + 900)}{1.2} = 881.5789 \text{ ml}$$

$$\text{Molarity} = \frac{2.6315 \times 1000}{881.5789} = 2.9849$$

$$\text{Molarity} = 2.98 \text{ M}$$

83. The number of significant figures in  $50000.020 \times 10^{-3}$  is .....

- A) 5      B) 8      C) 2      D) 10

**Solution : (Correct Answer: B )**

$$50000.020 \times 10^{-3}$$

84. A sample of a hydrate of barium chloride weighing 61 g was heated until all the water of hydration is removed. The dried sample weight is 52 g. The formula of the hydrated salt is : ( atomic mass,  $Ba = 137 \text{ amu}$ ,  $Cl = 35.5 \text{ amu}$  )

- A)  $BaCl_2 \cdot 4H_2O$   
B)  $BaCl_2 \cdot 3H_2O$   
C)  $BaCl_2 \cdot H_2O$   
D)  $BaCl_2 \cdot 2H_2O$

**Solution : (Correct Answer: D )**

Weight of hydrated  $BaCl_2 = 61 \text{ g}$

Weight of anhydrous  $BaCl_2 = 52 \text{ g}$

Loss in mass = 9 g

Assuming  $BaCl_2 \cdot xH_2O$  as hydrate mass of  $H_2O = 9 \text{ g}$

$$\text{Moles of } H_2O = \frac{9}{18} = 0.5$$

Grass molecular let of  $BaCl_2 = 208$

$$\% \text{ of } H_2O \text{ in this hydrated } BaCl_2 = \frac{9}{61} \times 100$$

$$= 14.75\%$$

$$= \frac{18x}{208+18x} \times 100 \text{ on solving } x = 2$$

This percentage is present in  $BaCl_2 \cdot 2H_2O$

85. The mass of carbon present in 0.5 mole of  $K_4[Fe(CN)_6]$  is ..... g

- A) 1.8      B) 18      C) 3.6      D) 36

**Solution : (Correct Answer: D )**

1 mole of  $K_4[Fe(CN)_6]$  contains 6 mole carbon i.e. 72 g

86. The molecular weight of hydrogen peroxide is 34. What is the unit of molecular weight

- A) g  
B) mol  
C)  $g\ mol^{-1}$   
D)  $mol\ g^{-1}$

**Solution : (Correct Answer: C )**

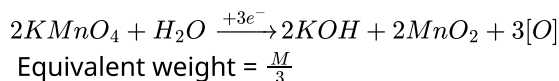
The molecular weight of hydrogen peroxide is 34. The unit of gram molecular weight is  $g/mol$ .

87. Which of the following statements is correct about equivalent weight of  $KMnO_4$

- A) It is one third of its molecular weight in alkaline medium  
B) It is one fifth of its molecular weight in alkaline medium  
C) It is equal to its molecular weight in acidic medium  
D) It is one third of its molecular weight in acidic medium

**Solution : (Correct Answer: A )**

(a) It is one third of its molecular weight in alkaline medium because it gives 3 nascent oxygen in alkaline medium.



88. The number of moles of sodium oxide in 620 g of it is ..... moles

- A) 1      B) 10  
C) 18      D) 100

**Solution : (Correct Answer: B )**

(b) Sodium oxide  $\rightarrow Na_2O$

Molecular weight =  $46 + 16 = 62$

62 gm of  $Na_2O = 1\ mole$

620 gm of  $Na_2O = 10\ mole$ .

89. The unit  $J\ Pa^{-1}$  is equivalent to

- A)  $m^3$       B)  $cm^3$   
C)  $dm^3$       D) None of these

**Solution : (Correct Answer: A )**

(a)  $J\ Pa^{-1}$ ; Unit of work is Joule and unit of pressure is Pascal.

Dimension of Joule i.e. work =  $F \times L = MLT^{-2} \times L$   
=  $[ML^2T^{-2}]$

$$\frac{1}{Pa} = \frac{1}{\text{Pressure}} = \frac{1}{\frac{F}{A}} = \frac{1 \times A}{F} = [MLT^{-1}]$$

$$\text{So, } J\ Pa^{-1} = [ML^2T^{-2}] = [L^2 \times L] = [L^3].$$

90. The significant figures in 3400 are

- A) 2      B) 5      C) 6      D) 4

**Solution : (Correct Answer: A )**

(a) As we know that all non zero unit are significant number. Therefore significant figure is 2.

## Biology - ( Botany )

91. A cell division in which a diploid somatic cell divides into two identical daughter cells is called

- A) Meiosis I      B) Meiosis II  
C) Mitosis      D) Cytokinesis

**Solution : (Correct Answer: C )**

Mitosis (equational division) occurs in somatic cells.

92. Match the following with respect to meiosis:

- |                |                     |
|----------------|---------------------|
| (a) Zygotene   | (i) Terminalization |
| (b) Pachytene  | (ii) Chiasmata      |
| (c) Diplotene  | (iii) Crossing over |
| (d) Diakinesis | (iv) Synapsis       |

Select the correct option from the following:

- (a) (b) (c) (d)

- A) (ii) (iv) (iii) (i)  
B) (iii) (iv) (i) (ii)  
C) (iv) (iii) (ii) (i)  
D) (i) (ii) (iv) (iii)

**Solution : (Correct Answer: C )**

- (iv) (iii) (ii) (i)

93. In which cells are biosynthetically active but they can not divided ?

- A) S phase      B)  $G_1$  phase  
C)  $G_0$  phase      D)  $G_2$  phase

**Solution : (Correct Answer: C )**

94. Bacterial cell divides every one minute. It takes 15 minutes a cup to be one-fourth full. How much time will it take to fill the cup?

- A) 30 minutes      B) 45 minutes  
C) 60 minutes      D) 17 minutes

**Solution : (Correct Answer: D )**

$$\frac{1}{4} \text{ of cup} = 15 \text{ minutes}$$

$$\frac{1}{2} \text{ of cup} = 16 \text{ minutes}$$

Cup of bacteria = 17 minutes

95. In which phase proteins for spindle fibre formation are synthesized

- A)  $G_1$  phase      B)  $G_2$  phase  
C) S-phase      D) Anaphase

**Solution : (Correct Answer: B )**

It's Obvious

96. Which of the following is not strictly a biomacromolecule?



- A) Proteins                      B) Lipids  
C) Polysaccharides          D) Nucleic acid

**Solution : (Correct Answer: B )**

Because the molecular weight of lipids does not exceed 800 Da but they come under the acid insoluble fraction.

97. In the plant 'ash', this is not present.

- A) Phosphate                      B) Sulphate  
C) Cellulose                      D) All are present

**Solution : (Correct Answer: C )**

98. About 98 percent of the mass of every living organism is composed of just six elements including carbon, hydrogen, nitrogen, oxygen and

- A) sulphur and magnesium  
B) magnesium and sodium  
C) calcium and phosphorus  
D) phosphorus and sulphur.

**Solution : (Correct Answer: D )**

(d) : Living organisms requires 6 elements in relatively large amounts. C, H, O, N, P, S. These elements contribute to the structural organization of living organisms.

99. The four elements that make up 99% of all elements found in a living system are

- A) *H, O, C, N*                      B) *C, H, O, S*  
C) *C, H, O, P*                      D) *C, N, O, P*

**Solution : (Correct Answer: A )**

It's Obvious

100. Weight of protein is

- A)  $> 12000$                       B)  $< 6000$   
C)  $< 12000$                       D)  $600 - 3000$

**Solution : (Correct Answer: A )**

It's Obvious

101. Which of the following statement was not explained in the cell theory given jointly by Schleiden and Schwann?

- A) All living organism are composed of cells and their products  
B) Cell is the structural and functional unit of living organisms  
C) Formation of new cells  
D) None of the above

**Solution : (Correct Answer: C )**

102. Match column-I (scientists) with column-II (discovery) and select the correct option.

| Column-I        | Column-II                                              |
|-----------------|--------------------------------------------------------|
| A. Leeuwenhoek  | I. First saw and described a living cell               |
| B. Robert Brown | II. Presence of cell wall is unique to plant cells     |
| C. Schleiden    | III. Discovered the nucleus                            |
| D. Schwann      | IV. All plants are composed of different kind of cells |

A) *A – I; B – III; C – IV; D – II*

B) *A – I; B – III; C – II; D – IV*

C) *A – III; B – I; C – IV; D – II*

D) *A – I; B – IV; C – II; D – III*

**Solution : (Correct Answer: A )**

Leeuwenhoek - He is commonly known as "the Father of Microbiology", and considered to be the first microbiologist. First saw and described a living cell. Robert Brown was a Scottish botanist and palaeobotanist who made important contributions nucleus and cytoplasmic streaming. Schleiden was a German botanist and co-founder of the cell theory, along with Theodor Schwann and Rudolf Virchow. He discovered that all plants are composed of different kind of cells. Schwann - He discovered that presence of cell wall is unique to plant cells.

103. The cell theory was given in year 1839 by Schleiden and Schwann. According to this theory all organisms are composed of cell and cells are the basic unit of life. How did this theory help in the field of science?

- A) It helped to study the working of cells.  
B) It helped in curing diseases caused by cell.  
C) It helped in restating the earlier theories on cell.  
D) It helped in introducing the use of microscopes to study cell.

**Solution : (Correct Answer: A )**

Cell theory explains the general principles of construction for all living things by focusing on studying the working of cells. The use of microscopes to study cell was in practice way before the cell theory was formulated.

104. Which of the following best illustrates "feedback" in development?

- A) Tissue *X* secretes *RNA* which changes the development of tissue *Y*.  
B) As tissue *X* develops, it secretes enzymes that inhibit the development of tissue *Y*.  
C) As tissue *X* develops, it secretes something that induces tissue *Y* to develop.  
D) As tissue *X* develops, it secretes something that slows down the growth of tissue *Y*.

**Solution : (Correct Answer: C )**



(c) : As tissue  $X$  develops it secretes something that induces tissue  $Y$  to develop indicating positive feedback mechanism.

105. ....examined a large number of .....and observed that all are composed of different kinds of cells which from the tissues

- A) Anton Von leuwenhoek, live cell
- B) Robert brown,nucleus
- C) Matthis schleiden, plant
- D) Theodore and animal

**Solution : (Correct Answer: C )**

106. Bark refers to all tissues exterior to the

- A) Cork cambium      B) Pericycle
- C) Vascular cambium      D) Periderm

**Solution : (Correct Answer: C )**

Bark is non technical term given to all tissues down the bast *i. e.*, upto secondary phloem.

107. Meristem is characterized by

- A) Isodiametric cells with cellulosic thin wall
- B) Absence of intercellular space and vacuole
- C) Absence of reserve food material and plastids
- D) All of these

**Solution : (Correct Answer: D )**

Meristems - Actively dividing cells.

108. In sclerenchymatous cells, the lignin deposition

- A) Is sometimes absent
- B) Begin from primary to secondary walls
- C) Begin from secondary to primary
- D) Is confined to middle lamella

**Solution : (Correct Answer: B )**

It's obvious.

109. The meristem in the root is

- A) Terminal
- B) Sub-apical or sub-terminal
- C) Intercalary
- D) Absent

**Solution : (Correct Answer: B )**

(b) In the roots the meristem is sub-terminal due to the presence of terminal cap.

110. A group of cells similar in form, function and origin is called

- A) Organ      B) Organella
- C) Tissue      D) None of these

**Solution : (Correct Answer: C )**

It's obvious.

111. Which of the following is not the lateral branches of the roots?

- A) Tertiary roots
- B) Secondary roots
- C) Primary root

D) More than one option is correct

**Solution : (Correct Answer: C )**

Primary root is not the lateral branches of the roots.

112. Androeium is epipetalous in this family

- A) Solanaceae      B) Brassicaceae
- C) Liliaceae      D) None

**Solution : (Correct Answer: A )**

113. The calyptrogen of the root apex forms

- A) Rhizoids      B) Root nodule
- C) Root hairs      D) Root cap

**Solution : (Correct Answer: D )**

(d) Root cap : In root apex, calyptrogen is covered by a cap like structure known as root cap, which helps the root apex from friction of soil.

114. Aestivation of corolla in Pea is

- A) Contorted
- B) Valvate
- C) Imbricate
- D) Vexillary

**Solution : (Correct Answer: D )**

(d) Vexillary aestivation : The posterior large standard petal overlaps the lateral petals called wings or,

alae on the margins while the lateral wings overlap the anterior keel or carinal petals, e.g., Pea, Gram.

115. Monocot plants are characterised by the presence of

- A) Tap roots      B) Fibrous roots
- C) Annulated roots      D) Stilt roots

**Solution : (Correct Answer: B )**

It's obvious.

116. Choose the correct statements about protonema

- A) Juvenile stage of moss is protonema
- B) It consists of slender; green, branching system of filaments
- C) Develops directly from a spore
- D) All of the above

**Solution : (Correct Answer: D )**

Juvenile stage of moss is protonema, which develops directly from a spore. It is a creeping, green, slender, branched and frequently filamentous stage

117. Kingdom-Plantae includes

- A) Algae, bryophytes and pteridophytes
- B) Algae, bryophytes, pteridophytes, gymnosperms and angiosperms
- C) Algae, fungi, pteridophytes, gymnosperms and angiosperms
- D) Algae, pteridophytes, gymnosperms and angiosperms

**Solution : (Correct Answer: B )**

The kingdom-Plantae includes algae, bryophytes, pteridophytes, gymnosperms and flowering plants (angiosperms). They are common on land, on sea shore and in freshwater

**118.** Karyotaxonomy is the modern branch of classification which is based on

- A) Number of chromosomes
- B) Bands found on chromosomes**
- C) Organic evolution
- D) Trinomial nomenclature

**Solution : (Correct Answer: B )**

It's obvious.

**119.** Which of the following form contain algae

- A) Equisetum
- B) Selaginella
- C) Marsilea
- D) None of these**

**Solution : (Correct Answer: D )**

(d) These are pteridophytes.

**120.** Natural system of classification differs from artificial system in

- A) Employing only one floral trait
- B) Taking only one vegetative trait
- C) Bringing out similarities and dissimilarities**
- D) Developing evolutionary trends

**Solution : (Correct Answer: C )**

It's Obvious.

**121.** Genus *Penicillium* belong to the class

- A) Basidiomycetes
- B) Ascomycetes**
- C) Phycomycetes
- D) Deuteromycetes

**Solution : (Correct Answer: B )**

It's Obvious

**122.** Which of the following is a nitrogen fixer

- A) *Ulothrix*
- B) *Anabaena***
- C) *Ulva*
- D) *Hydrodictyon*

**Solution : (Correct Answer: B )**

It's obvious.

**123.** Latest classification of biological kingdoms has been proposed by

- A) Linnaeus
- B) Haeckel
- C) Whittaker**
- D) John Ray

**Solution : (Correct Answer: C )**

It's Obvious.

**124.** Phenetic classification of organisms is based on

- A) Dendrogram based on *DNA* characteristics
- B) Sexual characteristics
- C) Observable characteristics of existing organisms**
- D) The ancestral lineage of existing organisms

**Solution : (Correct Answer: C )**

It's Obvious.

**125.** The scientist who created the group Protista for both unicellular plants and animals is

- A) Haeckel**
- B) Pasteur
- C) Lister
- D) Koch

**Solution : (Correct Answer: A )**

(a) A German biologist Ernst Haeckel created a third kingdom protista in 1866 for unicellular organism such as bacteria, protozoans, Algae and fungi.

**126.** Delete odd one for wheat.

- A) Poaceae
- B) Monocotyledonae
- C) Angiospermae
- D) Dicotyledonae**

**Solution : (Correct Answer: D )**

**127.** Which of the following taxonomic category of housefly is incorrectly matched?

- A) Genus - *Musca*
- B) Family - Muscidae
- C) Order - Primata**
- D) Class - Insecta

**Solution : (Correct Answer: C )**

Housefly, *Musca domestica*, is a fly of the order Diptera. It is the most common of all domestic flies. Primate is a mammal of the order primata.

**128.** Potato and brinjal belong to the genus *Solanum*, which reflects that

- A) They belong to single species
- B) They are a group of related species**
- C) They both are morphologically and structurally similar to each other in all respects
- D) They can always produce fertile hybrid

**Solution : (Correct Answer: B )**

Potato and Brinjal are a group of related species.

**129.** Which of the following is incorrect for reproduction?

- A) Unicellular organisms reproduce by cell division
- B) Reproduction is a characteristic of all living organisms**
- C) In unicellular organisms, reproduction and growth are linked together
- D) Non-living objects are incapable of reproducing

**Solution : (Correct Answer: B )**

Reproduction is absent in sterile organism like mule, hinny, sterile/infertile human couples, worker bees etc.

**130.** The fungi, the filamentous algae, the protonema of mosses, all easily multiply by \_\_\_\_.

- A) budding
- B) fission
- C) regeneration
- D) fragmentation**

**Solution : (Correct Answer: D )**

Fragmentation is a sexual mode of reproduction in which an organism is split into fragments. Each of these fragments develops into mature, fully grown individuals that are clones of the original organism.

The fungi, the filamentous algae and the protonema of mosses all easily multiply by fragmentation.

131. Which of the following are unique features of living organisms?

- A) Growth and reproduction
- B) Reproduction and ability to sense environment
- C) Metabolism and interaction
- D) All of the above

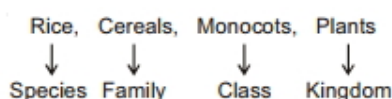
**Solution : (Correct Answer: D )**

All living organisms share certain unified and basic characteristics (including energy utilization, regulation or homeostasis, growth, development, reproduction, adaptation) metabolism and interaction.

132. Rice, cereals, monocots and plants represent

- A) Different taxa at different level
- B) Same taxa of different category
- C) Different category of same taxa
- D) Same category for different taxa

**Solution : (Correct Answer: A )**



133. In a hierarchical system of plant classification, which one of the following taxonomic ranks generally ends in 'ceae'?

- A) Family
- B) Genus
- C) Order
- D) Class

**Solution : (Correct Answer: A )**

(a) Family is a taxonomic category between the division and order. It includes one or more genera. Its suffix is 'aceae'.

134. A scientist having made significant contribution in the field of classification is

- A) Pasteur
- B) Oparin
- C) Darwin
- D) Linnaeus

**Solution : (Correct Answer: D )**

It's Obvious.

135. In Botanical nomenclature of plants

- A) Genus is written after the species
- B) Both in genus and species the first letter is a capital letter
- C) Genus and species may be same name
- D) Both genus and species are printed in italics

**Solution : (Correct Answer: D )**

(d) In botanical nomenclature of plants generic name always being with capital letter and the specific name with a small letter and printed with italics.

136. Match List I with List II.

| List I                         | List II                           |
|--------------------------------|-----------------------------------|
| A. Mast cells                  | I. Ciliated epithelium            |
| B. Inner surface of bronchiole | II. Areolar connective tissue     |
| C. Blood                       | III. Cuboidal epithelium          |
| D. Tubular parts of nephron    | IV. Specialised connective tissue |

Choose the correct answer from the options given below:

- A) A-III, B-IV, C-II, D-I
- B) A-I, B-II, C-IV, D-III
- C) A-II, B-III, C-I, D-IV
- D) A-II, B-I, C-IV, D-III

**Solution : (Correct Answer: D )**

Option (4) is the correct answer because,

- Areolar connective tissue contains fibroblasts (cells that produce and secrete fibres), macrophages and mast cells.

- Inner surface of bronchioles is lined by ciliated epithelium.

- Blood is a specialised connective tissue.

- Tubular parts of nephron are lined by cuboidal epithelium.

137. Match the following:

| Column - I              | Column - II                 |
|-------------------------|-----------------------------|
| A. Caecum in earthworm  | 1. Tubular                  |
| B. Caecum in cockroach  | 2. Forgut                   |
| C. Gizzard in earthworm | 3. 26 <sup>th</sup> segment |
| D. Gizzard in cockroach | 4. 8 - 9 segment            |

- A) A - 1, B - 4, C - 2, D - 3
- B) A - 4, B - 2, C - 1, D - 3
- C) A - 3, B - 1, C - 4, D - 2
- D) A - 3, B - 1, C - 2, D - 3

**Solution : (Correct Answer: C )**

138. Which of the following characteristics is incorrect with respect to cockroach?

- A) A ring of gastric caeca is present at the junction of midgut and hind gut.
- B) Hypopharynx lies within the cavity enclosed by the mouth parts.
- C) In females, 7<sup>th</sup> - 9<sup>th</sup> sterna together form a genital pouch.
- D) 10<sup>th</sup> abdominal segment in both sexes, bears a pair of anal cerci.

**Solution : (Correct Answer: A )**

139. Consider the following statements regarding cockroach and mark the correct option.

- (i) Head is formed by the fusion of 6 - segments.

- (ii) Mouth parts are biting and chewing type.  
(iii) Crop is the part of mid gut.

- A) Only (ii) is correct  
B) (ii) and (iii) are correct  
C) (i) and (ii) are correct  
D) Only (iii) is correct

**Solution : (Correct Answer: C )**

Crop is the part of foregut which is used for storing of food.

140. Lack of blood supply and presence of the noncellular basement membrane are the characteristics of the

- A) muscular tissue  
B) fluid connective tissue  
C) epithelial tissue  
D) nervous tissue

**Solution : (Correct Answer: C )**

Epithelial tissue covers the body surface, or lines the body cavity. The cells rest upon a non-cellular basement membrane which is secreted by epithelial cells. It is not nourished by blood, so nonvascularized.

141. Match the following columns.

| Column – I        | Column – II                        |
|-------------------|------------------------------------|
| A. Areolar tissue | 1. Fat cells                       |
| B. Adipose tissue | 2. Osteocytes                      |
| C. Ligament       | 3. Loose connective tissue         |
| D. Bone           | 4. Dense regular connective tissue |

- A) A – 3, B – 1, C – 4, D – 2  
B) A – 1, B – 2, C – 3, D – 4  
C) A – 4, B – 3, C – 2, D – 1  
D) A – 2, B – 1, C – 4, D – 3

**Solution : (Correct Answer: A )**

142. A : Intercellular material is minimum between the cells of epithelial tissue.

R : Epithelial cells are not secretory in nature.

- A) Assertion and Reason both are correct and also correct explanation.  
B) Assertion and Reason both are correct but not explanation of assertion.  
C) Assertion is correct, but Reason is incorrect.  
D) Both Assertion and Reason are incorrect.

**Solution : (Correct Answer: C )**

Assertion is true as epithelial tissue consist of variously shaped cells closely arranged in one or more layers. There is little intercellular material between epithelial cells.

Reason is false as epithelial cells are secretory in nature, epithelial cells secrete basal lamina (basement membrane)

Assertion is true but reason is false.

143. Mark the correct option with respect to tissue, location, and its function.

| Tissue          | Location | Function  |
|-----------------|----------|-----------|
| Reticular issue | Spleen   | Secretion |

| Tissue                             | Location | Function     |
|------------------------------------|----------|--------------|
| Brush bordered cuboidal epithelium | PCT      | Reabsorption |

| Tissue                  | Location              | Function                                                          |
|-------------------------|-----------------------|-------------------------------------------------------------------|
| Neurosensory epithelium | Taste buds and cornea | Conversion of all types of electrical stimuli to chemical stimuli |

| Tissue               | Location                | Function                 |
|----------------------|-------------------------|--------------------------|
| Glandular epithelium | Lining of blood vessels | To handle blood pressure |

**Solution : (Correct Answer: B )**

| Tissue                             | Location | Function     |
|------------------------------------|----------|--------------|
| Brush bordered cuboidal epithelium | PCT      | Reabsorption |

Reticular tissue helps in filtering blood, removing antigens, and old RBCs in the spleen. The cornea has no sensory epithelium. Simple Squamous epithelium lines the blood vessels.

144. Non-ciliated simple columnar epithelium often contains which increase the surface area for secretion and absorption.

- A) flagella  
B) collagen fibres  
C) microvilli  
D) all of these

**Solution : (Correct Answer: C )**

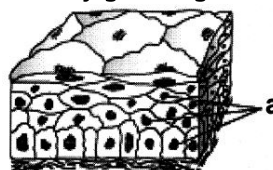
The columnar epithelium is composed of a single layer of tall and slender cells. Their free surface have microvilli found in the lining of stomach and intestine and help in secretion and absorption.

145. Body cavity of cockroach is also called

- A) Blood cavity  
B) Genital chamber  
C) Heart chamber  
D) Cloaca

**Solution : (Correct Answer: A )**

146. Identify given figure and indicated part of 'a'.



- A) Ciliated epithelium - nucleus  
B) Compound epithelium -multilayered cells  
C) Compound epithelium - multicellular gland  
D) Ciliated epithelium -multilayered cells.

**Solution : (Correct Answer: B )**

147. What external changes are visible after the last moult of a cockroach nymph?

- A) Both forewings and hindwings develop
- B) Labium develops
- C) Mandibles become harder
- D) Anal cerci develop

**Solution : (Correct Answer: A )**

(a) : Cockroach undergoes paurometabolous development. The nymph moults about 6– 7 times to reach the adult form. The next to last nymphal stage has wing pads but only adult cockroaches have wings.

148. Muscular tissue is differentiated into

- A) Unstriated, striped
- B) Striped, cardiac
- C) Cardiac muscle, unstriated
- D) Unstriated, striated and cardiac

**Solution : (Correct Answer: D )**

It's Obvious

149. Identify the tissue shown in the diagram and match with its characteristics and its location.



- A) Smooth muscles, show branching, found in the wall of the heart
- B) Cardiac muscles, unbranched muscles, found in the walls of the heart
- C) Striated muscles, tapering at both ends, attached with the bones of the ribs
- D) Skeletal muscles show striations and are closely attached with the bones of the limbs

**Solution : (Correct Answer: D )**

(d) : Locomotion (performed by limbs) in humans depends on the movements of muscle fibres. Skeletal muscles are attached to the bones by tendons and help in the movement of the parts of skeleton. These muscles are under the control of conscious mind and are called voluntary muscles. Under the microscope, these muscles show transverse stripes and hence are designated as striated muscles.

150. In frog, respiration occurs by

- A) Lungs
- B) Trachae
- C) Gills only
- D) Both (a) and (b)

**Solution : (Correct Answer: A )**

It's Obvious

151. Which of the following is not a median part in the brain cavity of frog

- A) Iter
- B) Diocoel
- C) Metacoel
- D) Optocoel

**Solution : (Correct Answer: D )**

(d) Optocoels are paired cavities found in the optic lobes of frog. These open into iter, the canal in midbrain.

152. Ninth cranial nerve of frog is

- A) Trochlear
- B) Trigeminal
- C) Oculomotor
- D) Glossopharyngeal

**Solution : (Correct Answer: D )**

(d) Glossopharyngeal is ninth cranial nerve of frog, it supplies the taste buds of tongue and muscles of the pharynx. Some fibres carry impulses from tongue,

while the other fibres are responsible for pharynx movement as they are concerned with swallowing reflex.

153. Inner surface of fallopian tubes, bronchi and bronchioles are lined by

- A) Squamous epithelium
- B) Ciliated epithelium
- C) Columnar epithelium
- D) Cubical epithelium

**Solution : (Correct Answer: B )**

It's Obvious

154. Frog is

- A) Arboreal
- B) Terrestrial
- C) Fully aquatic
- D) Both aquatic and terrestrial

**Solution : (Correct Answer: D )**

(d) Frog is amphibious or stenohygrous,

*i. e.*, lives in or near fresh water bodies, like ponds, lakes, ditches, marshes and even slow running waters.

155. The intestine and stomach in mammals are lined by

- A) Cuboidal epithelium
- B) Columnar epithelium
- C) Squamous epithelium
- D) Stratified epithelium

**Solution : (Correct Answer: B )**

(b) Simple columnar epithelium lines the stomach, the small and large intestines, the digestive glands and the gall bladder.

156. Growth of young cartilage takes place by

- A) Division of young chondrocytes
- B) Formation of more intercellular substance



C) Deposition of new layer of cartilage at its surface

D) All of the above

**Solution : (Correct Answer: D )**

It's Obvious.

157. External ear (*pinna*) contains a hard, flexible structure composed of

A) Bone

B) Cartilage

C) Tendon

D) Ligament

**Solution : (Correct Answer: B )**

(b) Due to presence of yellow fibre, elastic cartilage is very flexible. It is found in epiglottis, pinna of ear, wall of eustachian tube and at the tip of nose.

158. In closed circulatory system

A) The cells and tissues are directly bathed in the blood pumped out by heart

B) Arteries and veins are lacking

C) The capillaries are largest blood vessels and closed at their ends

D) Blood circulates through a series of vessels of varying diameters

**Solution : (Correct Answer: D )**

Closed circulatory system is a type of blood vascular system in which blood flows inside blood vessels of varying diameter (arteries, veins, capillaries) without coming in direct contact with body cells. Closed circulatory system is observed in annelids, cephalopod molluscs, chordates.

159. The only incorrectly matched pair is

| Phylum | Level of organisation |
|--------|-----------------------|
|--------|-----------------------|

A) Porifera - Cellular level

B) Cnidaria - Tissue level

C) Annelida - Organ level only

D) Mollusca - Organ system level

**Solution : (Correct Answer: C )**

Porifera - Cellular level

Cnidaria - Tissue level

Annelida - Organ-system level

Mollusca - Organ-system level

160. Select the incorrect option with respect to features present in three phylum

Characters - Arthropoda - Mollusca - Chordata

A) Coelom - Absent - Absent - Absent

B) Segmentation - Present - Absent - Present

C) Digestive system - Complete - Complete - Complete

D) Circulatory system - Present - Present - Present

**Solution : (Correct Answer: A )**

161. Three types of body cavity are

A) True coelom, pseudocoelom and haemocoel

B) Pseudocoelom, protocoelom and acoelom

C) Acoelom, deuterocoel and homocoel

D) Protocoel, deuterocoel and pseudocoelom

**Solution : (Correct Answer: A )**

Three types of body cavity are true coelom, pseudocoelom and haemocoel. In phylum - Arthropoda and Mollusca a haemocoel is seen, the true coelom is reduced and blood fills the spaces between the viscera

162. Animals are classified on the basis of which of the following features?

I. Coelomic cavity II. Level of organisation III. Notochord IV. Skeletal structure

A) I and II

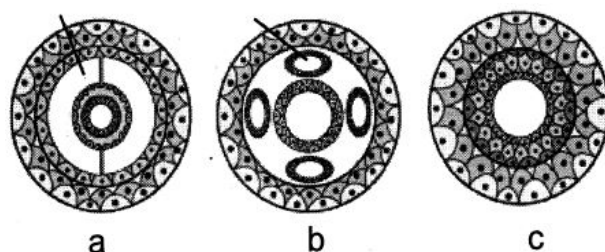
B) I and III

C) I, II and III

D) II and IV

**Solution : (Correct Answer: C )**

163. According to given figure a, b, c choose correct answer for coelomate



A) a - Aschelminthes, b - Platyhelminthes c - arthropods

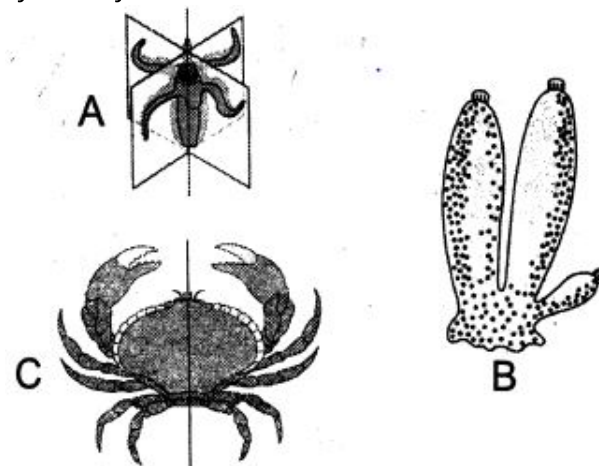
B) a - arthropods, b - Platyhelminthes c - Aschelminthes

C) a - arthropods, b - Aschelminthes c - Platyhelminthes

D) a - Platyhelminthes, b - Aschelminthes c - arthropods

**Solution : (Correct Answer: C )**

164. Which one of the following figure shows radial symmetry



A) A

B) B

C) C

D) Give all

**Solution : (Correct Answer: A )**



165. Choose correct pair :

class - Character

- A) Mammal - Poikilotherms
- B) Reptile - Homoiothermous
- C) Amphibians - Oviparous and external fertilization
- D) Aves - Three chambered heart

**Solution : (Correct Answer: C )**

166. Metameric segmentation is the characteristic of

- A) mollusca and chordata
- B) platyhelminthes and arthropoda
- C) echinodermata and annelida
- D) annelida and arthropoda.

**Solution : (Correct Answer: D )**

(d) : The term metamerism refers to a linear repetition of parts in an animal body. It occurs in three highly organized phyla : Annelida, Arthropoda and Chordata. Each segment is called a metamere, or somite. Segmentation often affects both external and internal structures. Such a condition is called metameric segmentation. In chordates, the segmentation is apparent only in the embryonic stage. In the adult chordates, segmentation is visible in the internal structures, such as vertebrae, ribs, nerves and blood vessels. Other animals have unsegmented bodies.

167. The term for body cavity present in triploblastic animals is

- A) Haemocoel
- B) Pseudocoelom
- C) Coelom
- D) All the above

**Solution : (Correct Answer: C )**

(c) Triploblastic animals have three layers - the ectoderm, mesoderm and endoderm. The true coelom is a body cavity which arises as a cavity in embryonic mesoderm.

168. In which of the following jaws are found

- A) Herdmania
- B) Fish
- C) Petromyzon
- D) Amphioxus

**Solution : (Correct Answer: B )**

(b) On the basis of presence or absence of jaw subphylum vertebrate is classified into Agnathostomata and Gnathostomata. In agnathostomata all those animals are included in which jaw is present.

169. The first true organ system grade body organisation is developed in

- A) annelida
- B) Coelenterata
- C) Platyhelminthes
- D) Arthropoda

**Solution : (Correct Answer: A )**

It's obvious.

170. The haemocoel of an insect is actually a

- A) Modified blood vessel
- B) True coelom
- C) Pseudocoelom
- D) Schizocoelom

**Solution : (Correct Answer: B )**

It's obvious

171. Leech secretes which of the following anticoagulant

- A) Hirudin
- B) Heparin
- C) Serotonin
- D) Histamine

**Solution : (Correct Answer: A )**

It's obvious.

172. All worms are

- A) Triploblastic
- B) Segmented
- C) Endo-parasites
- D) Free-living

**Solution : (Correct Answer: A )**

It's obvious.

173. A true coelom is absent in phylum

- A) Nematoda
- B) Annelida
- C) Echinodermata
- D) Mollusca

**Solution : (Correct Answer: A )**

(a) In nemathelminthes or round worm, coelom is present but it is derived from blastocoel. It is not lined by peritonium. This type of coelom is known as pseudocoelom.

174. Phylum coelenterata has remained at

- A) Cellular level of organisation
- B) Organ level of organisation
- C) Tissue level of organisation
- D) Organ system level of organisation

**Solution : (Correct Answer: C )**

It's obvious.

175. Which one is not typical to all porifers

- A) Perforated body
- B) Choanocytes
- C) System of pores and canal
- D) Presence of spongin fibres

**Solution : (Correct Answer: D )**

(d) Porifers possess skeleton of calcareous or siliceous spicules or of spongin fibres or of both or no skeleton.

176. Body cavity lined by mesoderm is called

- A) Coelenteron
- B) Pseudocoel
- C) Coelom
- D) Blastocoel

**Solution : (Correct Answer: C )**

It's obvious.

177. Body does not show any segmentation in

- A) Frog
- B) Cockroach

C) Earthworm      D) Star Fish/Hydra

**Solution : (Correct Answer: D )**

It's obvious.

178. Hydra is

A) Coral    B) Worm    C) Polyp    D) Medusa

**Solution : (Correct Answer: C )**

(c) Hydra is a polyp-like or polypoid coelenterate with a tubular or cylindrical body.

179. Which group is commonly called as 'sea stick'

A) Arthropoda      B) Echinodermata  
C) Coelenterata      D) Porifera

**Solution : (Correct Answer: C )**

It's obvious.

180. Level of organisation in sponges is

A) Cellular level  
B) Acellular level  
C) Tissue level  
D) Organ-system level

**Solution : (Correct Answer: A )**

(a) Sponges are "multicellular grade" animals but exhibit cellular level of organization.